Задание 1

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: liveness-ex
spec:
  selector:
    matchLabels:
      app: liveness-ex
  template:
    metadata:
      labels:
        app: liveness-ex
    spec:
      containers:
      - name: exec-container
        image: busybox
        command: ["sh", "-c", "touch /tmp/healthy; sleep 3600"]
        livenessProbe:
          exec:
           command:
           - cat
           - /tmp/healthy
          initialDelaySeconds: 5
          periodSeconds: 5

      - name: http-container
        image: nginx:1.25
        ports:
        - containerPort: 80
        livenessProbe:
          httpGet:
            path: /healthz
            port: 80
            httpHeaders:
            - name: custom-header
              value: health-check
          initialDelaySeconds: 5
```

```yaml
          periodSeconds: 5
          timeoutSeconds: 5

      - name: tcp-container
        image: redis:7-alpine
        ports:
        - containerPort: 6379
        livenessProbe:
          tcpSocket:
              port: 6379
          initialDelaySeconds: 5
          periodSeconds: 5
```

deploy.yaml

```
lly@lly ~/m/s/t/1 (master)> kubectl get pods -l app=liveness-ex
NAME                                READY   STATUS    RESTARTS   AGE
liveness-ex-5ddd8b7648-qwm7t        3/3     Running   0          5s
```

kubectl get pods -l app=liveness-ex

```
Events:
  Type     Reason     Age                   From               Message
  ----     ------     ----                  ----               -------
  Normal   Scheduled  5m14s                 default-scheduler  Successfully assigned default/liveness-ex-5ddd8b7648-qwm7t to minikube
  Normal   Pulling    5m14s                 kubelet            spec.containers{exec-container}: Pulling image "busybox"
  Normal   Pulled     5m12s                 kubelet            spec.containers{exec-container}: Successfully pulled image "busybox" in 2.007s (2.007s including waiting
). Image size: 4421262 bytes.
  Normal   Created    5m12s                 kubelet            spec.containers{exec-container}: Container created
  Normal   Started    5m12s                 kubelet            spec.containers{exec-container}: Container started
  Normal   Created    5m11s                 kubelet            spec.containers{tcp-container}: Container created
  Normal   Pulled     5m11s                 kubelet            spec.containers{tcp-container}: Container image "redis:7-alpine" already present on machine and can be a
ccessed by the pod
  Normal   Started    5m11s                 kubelet            spec.containers{tcp-container}: Container started
  Normal   Started    4m24s (x4 over 5m11s) kubelet            spec.containers{http-container}: Container started
  Normal   Killing    4m9s (x4 over 4m54s)  kubelet            spec.containers{http-container}: Container http-container failed liveness probe, will be restarted
  Normal   Created    3m39s (x5 over 5m11s) kubelet            spec.containers{http-container}: Container created
  Normal   Pulled     3m39s (x5 over 5m12s) kubelet            spec.containers{http-container}: Container image "nginx:1.25" already present on machine and can be acce
ssed by the pod
  Warning  BackOff    3m8s (x4 over 4m9s)   kubelet            spec.containers{http-container}: Back-off restarting failed container http-container in pod liveness-ex-
5ddd8b7648-qwm7t_default(99f23bed-19f8-4dda-8c07-a98b191c4ce2)
  Warning  Unhealthy  89s (x21 over 5m4s)   kubelet            spec.containers{http-container}: Liveness probe failed: HTTP probe failed with statuscode: 404
```

kubectl describe pods -l app=liveness-ex

```
lly@lly ~/m/s/t/1 (master)> kubectl get pods -l app=liveness-ex --watch
NAME                                READY   STATUS             RESTARTS        AGE
liveness-ex-5ddd8b7648-qwm7t        2/3     CrashLoopBackOff   6 (2m44s ago)   6m29s
liveness-ex-5ddd8b7648-qwm7t        3/3     Running            7 (2m50s ago)   6m35s
liveness-ex-5ddd8b7648-qwm7t        2/3     CrashLoopBackOff   7 (1s ago)      6m51s
```

kubectl get pods -l app=liveness-ex –watch

```
lly@lly ~/m/s/t/1 (master)> kubectl logs liveness-ex-5ddd8b7648-qwm7t http-container
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/27 08:37:07 [notice] 1#1: using the "epoll" event method
2026/03/27 08:37:07 [notice] 1#1: nginx/1.25.5
2026/03/27 08:37:07 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2026/03/27 08:37:07 [notice] 1#1: OS: Linux 6.6.95
2026/03/27 08:37:07 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/27 08:37:07 [notice] 1#1: start worker processes
2026/03/27 08:37:07 [notice] 1#1: start worker process 29
2026/03/27 08:37:07 [notice] 1#1: start worker process 30
10.244.0.1 - - [27/Mar/2026:08:37:13 +0000] "GET /healthz HTTP/1.1" 404 153 "-" "kube-probe/1.35" "-"
2026/03/27 08:37:13 [error] 29#29: *1 open() "/usr/share/nginx/html/healthz" failed (2: No such file or directory), client: 10.244.0.1, server: localhost, request: "GET
 /healthz HTTP/1.1", host: "10.244.0.11:80"
10.244.0.1 - - [27/Mar/2026:08:37:18 +0000] "GET /healthz HTTP/1.1" 404 153 "-" "kube-probe/1.35" "-"
2026/03/27 08:37:18 [error] 30#30: *2 open() "/usr/share/nginx/html/healthz" failed (2: No such file or directory), client: 10.244.0.1, server: localhost, request: "GET
 /healthz HTTP/1.1", host: "10.244.0.11:80"
2026/03/27 08:37:23 [error] 29#29: *3 open() "/usr/share/nginx/html/healthz" failed (2: No such file or directory), client: 10.244.0.1, server: localhost, request: "GET
 /healthz HTTP/1.1", host: "10.244.0.11:80"
10.244.0.1 - - [27/Mar/2026:08:37:23 +0000] "GET /healthz HTTP/1.1" 404 153 "-" "kube-probe/1.35" "-"
2026/03/27 08:37:23 [notice] 1#1: signal 3 (SIGQUIT) received, shutting down
2026/03/27 08:37:23 [notice] 29#29: gracefully shutting down
2026/03/27 08:37:23 [notice] 30#30: gracefully shutting down
2026/03/27 08:37:23 [notice] 29#29: exiting
2026/03/27 08:37:23 [notice] 30#30: exiting
2026/03/27 08:37:23 [notice] 30#30: exit
2026/03/27 08:37:23 [notice] 29#29: exit
2026/03/27 08:37:23 [notice] 1#1: signal 17 (SIGCHLD) received from 29
2026/03/27 08:37:23 [notice] 1#1: worker process 29 exited with code 0
2026/03/27 08:37:23 [notice] 1#1: worker process 30 exited with code 0
2026/03/27 08:37:23 [notice] 1#1: exit
```

kubectl logs <pod_name> http-container

```
lly@lly ~/m/s/t/1 (master)> kubectl get pods -l app=liveness-ex
NAME                              READY   STATUS             RESTARTS        AGE
liveness-ex-5ddd8b7648-qwm7t      2/3     CrashLoopBackOff   9 (2m13s ago)   14m
```

kubectl get pods -l app=liveness-ex (количество перезапусков)

Задание 2

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: combined
spec:
  selector:
    matchLabels:
      app: combined
  template:
    metadata:
      labels:
        app: combined
    spec:
      containers:
      - name: nginx-container
        image: nginx:1.25
        command: ["/bin/sh"]
        args:
        - "-c"
        - |
          sleep 40;
          nginx -g 'daemon off;';

        livenessProbe:
          httpGet:
            path: /
            port: 80
          initialDelaySeconds: 50
          periodSeconds: 10

        readinessProbe:
          httpGet:
            path: /
            port: 80
          initialDelaySeconds: 5
          periodSeconds: 5
        ports:
        - containerPort: 80
```

deploy.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: combined-svc
spec:
  selector:
    app: combined
  type: ClusterIP
  ports:
  - name: combined-svc
    protocol: TCP
    port: 80
    targetPort: 80
```

combined-svc.yaml (для проверки endpoints)

```
lly@lly ~/m/s/t/2 (master)> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
combined-5b44bd6646-qm48b     0/1     Running   0          5s
combined-5b44bd6646-qm48b     1/1     Running   0          42s
```

kubectl get pods -w

```
lly@lly ~/m/s/t/2 (master)> kubectl get endpoints -w
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSli
ce
NAME           ENDPOINTS             AGE
combined-svc                         12s
kubernetes     192.168.39.153:8443   37h
combined-svc   10.244.0.29:80        42s
```

kubectl get endpoints -w

```
  Containers:
   nginx-container:
    Image:      nginx:1.25
    Port:       80/TCP
    Host Port:  0/TCP
    Command:
      /bin/sh
    Args:
      -c
      sleep 40;
      nginx -g 'daemon off;';

    Liveness:      http-get http://:80/ delay=50s timeout=1s period=10s #success=1 #failure=3
    Readiness:     http-get http://:80/ delay=5s timeout=1s period=5s #success=1 #failure=3
```

kubectl describe deployment/combined (разница конфигурации Readiness and Liveness)

```
lly@lly ~/m/s/t/2 (master)> kubectl logs combined-5b44bd6646-qm48b
2026/03/27 09:44:26 [notice] 8#8: using the "epoll" event method
2026/03/27 09:44:26 [notice] 8#8: nginx/1.25.5
2026/03/27 09:44:26 [notice] 8#8: built by gcc 12.2.0 (Debian 12.2.0-14)
2026/03/27 09:44:26 [notice] 8#8: OS: Linux 6.6.95
2026/03/27 09:44:26 [notice] 8#8: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/27 09:44:26 [notice] 8#8: start worker processes
2026/03/27 09:44:26 [notice] 8#8: start worker process 9
2026/03/27 09:44:26 [notice] 8#8: start worker process 10
10.244.0.1 - - [27/Mar/2026:09:44:27 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:44:32 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:44:36 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:44:37 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:44:42 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:44:46 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:44:47 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:44:52 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:44:56 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:44:57 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:02 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:06 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:07 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:12 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:16 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:17 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:22 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:26 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:27 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:32 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:36 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:37 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:42 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:46 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:47 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:52 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:56 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:45:57 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:46:02 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
10.244.0.1 - - [27/Mar/2026:09:46:06 +0000] "GET / HTTP/1.1" 200 615 "-" "kube-probe/1.35" "-"
```

kubectl logs <pod_name>

```
State:          Running
  Started:      Fri, 27 Mar 2026 12:43:46 +0300
Ready:          True
```

kubectl describe pod <pod_name> (время перехода в состояние Ready)

```
Events:
  Type     Reason     Age                    From               Message
  ----     ------     ----                   ----               -------
  Normal   Scheduled  7m53s                  default-scheduler  Successfully assigned default/combined-5b44bd6646-qm48b to minikube
  Normal   Pulled     7m53s                  kubelet            spec.containers{nginx-container}: Container image "nginx:1.25" already present on machine and can be acc
essed by the pod
  Normal   Created    7m53s                  kubelet            spec.containers{nginx-container}: Container created
  Normal   Started    7m53s                  kubelet            spec.containers{nginx-container}: Container started
  Warning  Unhealthy  7m17s (x7 over 7m47s)  kubelet            spec.containers{nginx-container}: Readiness probe failed: Get "http://10.244.0.29:80/": dial tcp 10.244.
0.29:80: connect: connection refused
```

kubectl describe pod <pod_name> (анализ временной шкалы событий запуска)

Задание 3

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: volume
  labels:
    app: volume
spec:
  selector:
    matchLabels:
      app: volume
  template:
    metadata:
      labels:
        app: volume
    spec:
      containers:
      - name: generator
        image: alpine
        command: ["/bin/sh"]
        args: ["-c", "while true; do echo $(date) | tee -a /data/log.txt; sleep 10; done"]
        volumeMounts:
        - name: shared-data
          mountPath: /data

      - name: reader
        image: alpine
        command: ["/bin/sh"]
        args: ["-c", "tail -f /data/log.txt"]
        volumeMounts:
        - name: shared-data
          mountPath: /data

      volumes:
        - name: shared-data
          emptyDir: {}
```

deploy.yaml

```
lly@lly ~/m/s/t/3 (master)> kubectl get pods
NAME                      READY   STATUS    RESTARTS   AGE
volume-77f6d675f4-7dnvb   2/2     Running   0          20s
```

kubectl get pods

```
lly@lly ~/m/s/t/3 (master)> kubectl exec volume-77f6d675f4-7dnvb -c generator -- ls -i /data/
9437998 log.txt
lly@lly ~/m/s/t/3 (master)> kubectl exec volume-77f6d675f4-7dnvb -c reader -- ls -i /data/
9437998 log.txt
```

kubectl exec <pod_name> -c <container_name> -- ls -i /data/

```
lly@lly ~/m/s/t/3 (master)> kubectl logs volume-77f6d675f4-7dnvb -c generator
Fri Mar 27 11:17:21 UTC 2026
Fri Mar 27 11:17:31 UTC 2026
Fri Mar 27 11:17:41 UTC 2026
Fri Mar 27 11:17:51 UTC 2026
Fri Mar 27 11:18:01 UTC 2026
Fri Mar 27 11:18:11 UTC 2026
Fri Mar 27 11:18:21 UTC 2026
Fri Mar 27 11:18:31 UTC 2026
Fri Mar 27 11:18:41 UTC 2026
Fri Mar 27 11:18:51 UTC 2026
Fri Mar 27 11:19:01 UTC 2026
Fri Mar 27 11:19:11 UTC 2026
Fri Mar 27 11:19:21 UTC 2026
Fri Mar 27 11:19:31 UTC 2026
Fri Mar 27 11:19:41 UTC 2026
Fri Mar 27 11:19:51 UTC 2026
Fri Mar 27 11:20:01 UTC 2026
Fri Mar 27 11:20:11 UTC 2026
Fri Mar 27 11:20:21 UTC 2026
Fri Mar 27 11:20:31 UTC 2026
Fri Mar 27 11:20:41 UTC 2026
Fri Mar 27 11:20:51 UTC 2026
Fri Mar 27 11:21:01 UTC 2026
Fri Mar 27 11:21:11 UTC 2026
Fri Mar 27 11:21:21 UTC 2026
Fri Mar 27 11:21:31 UTC 2026
```

kubectl logs <pod_name> -c generator

```
lly@lly ~/m/s/t/3 (master)> kubectl logs volume-77f6d675f4-7dnvb -c reader
Fri Mar 27 11:17:21 UTC 2026
Fri Mar 27 11:17:31 UTC 2026
Fri Mar 27 11:17:41 UTC 2026
Fri Mar 27 11:17:51 UTC 2026
Fri Mar 27 11:18:01 UTC 2026
Fri Mar 27 11:18:11 UTC 2026
Fri Mar 27 11:18:21 UTC 2026
Fri Mar 27 11:18:31 UTC 2026
Fri Mar 27 11:18:41 UTC 2026
Fri Mar 27 11:18:51 UTC 2026
Fri Mar 27 11:19:01 UTC 2026
Fri Mar 27 11:19:11 UTC 2026
Fri Mar 27 11:19:21 UTC 2026
Fri Mar 27 11:19:31 UTC 2026
Fri Mar 27 11:19:41 UTC 2026
Fri Mar 27 11:19:51 UTC 2026
Fri Mar 27 11:20:01 UTC 2026
Fri Mar 27 11:20:11 UTC 2026
Fri Mar 27 11:20:21 UTC 2026
Fri Mar 27 11:20:31 UTC 2026
Fri Mar 27 11:20:41 UTC 2026
Fri Mar 27 11:20:51 UTC 2026
Fri Mar 27 11:21:01 UTC 2026
Fri Mar 27 11:21:11 UTC 2026
Fri Mar 27 11:21:21 UTC 2026
Fri Mar 27 11:21:31 UTC 2026
Fri Mar 27 11:21:41 UTC 2026
Fri Mar 27 11:21:51 UTC 2026
```

kubectl logs <pod_name> -c reader

Задание 4

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: timeouts
spec:
  selector:
    matchLabels:
      app: timeouts
  template:
    metadata:
      labels:
        app: timeouts
    spec:
      containers:
      - name: timeouts
        image: nginx:1.25
        command: ["/bin/sh"]
        args:
        - "-c"
        - |
          sleep 10;
          nginx -g 'daemon off;';

        livenessProbe:
          httpGet:
            port: 80
            path: /health
          initialDelaySeconds: 2
          timeoutSeconds: 1
          periodSeconds: 5

        readinessProbe:
          httpGet:
            path: /health
            port: 80
          initialDelaySeconds: 2
          timeoutSeconds: 1
          periodSeconds: 5
        ports:
        - containerPort: 80
```

timeouts.yaml

```
lly@lly ~/m/s/t/4 (master)> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
timeouts-75cfb4fc66-7x775     0/1     Running   0          1s
timeouts-75cfb4fc66-7x775     0/1     Running   1 (1s ago)   46s
timeouts-75cfb4fc66-7x775     0/1     Running   2 (1s ago)   91s
```

kubectl get pods -w

```
Containers:
  timeouts:
    Container ID:  docker://9712227e92bba4b41fcaa7a99614b44a267f11b704413eb103463a
32b9d1f9fa
    Image:         nginx:1.25
    Image ID:      docker-pullable://nginx@sha256:a484819eb60211f5299034ac80f6a681
b06f89e65866ce91f356ed7c72af059c
    Port:          80/TCP
    Host Port:     0/TCP
    Command:
      /bin/sh
    Args:
      -c
      sleep 10;
      nginx -g 'daemon off;';

    State:          Running
      Started:      Tue, 31 Mar 2026 17:28:28 +0300
    Last State:     Terminated
      Reason:       Error
      Exit Code:    137
      Started:      Tue, 31 Mar 2026 17:27:43 +0300
      Finished:     Tue, 31 Mar 2026 17:28:27 +0300
    Ready:          False
    Restart Count:  3
    Liveness:       http-get http://:80/health delay=2s timeout=1s period=5s #succ
ess=1 #failure=3
    Readiness:      http-get http://:80/health delay=2s timeout=1s period=5s #succ
ess=1 #failure=3
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-bc99n (ro
)
```

kubectl describe pod <pod_name>

```
Events:
  Type     Reason     Age                  From               Message
  ----     ------     ----                 ----               -------
  Normal   Scheduled  2m18s                default-scheduler  Successfully assigne
d default/timeouts-75cfb4fc66-7x775 to minikube
  Warning  Unhealthy  86s (x3 over 2m11s)  kubelet            spec.containers{time
outs}: Readiness probe failed: Get "http://10.244.0.54:80/health": dial tcp 10.244
.0.54:80: connect: connection refused
  Warning  Unhealthy  83s (x4 over 2m13s)  kubelet            spec.containers{time
outs}: Liveness probe failed: Get "http://10.244.0.54:80/health": dial tcp 10.244.
0.54:80: connect: connection refused
  Warning  Unhealthy  78s (x2 over 2m3s)   kubelet            spec.containers{time
outs}: Liveness probe failed: HTTP probe failed with statuscode: 404
  Warning  Unhealthy  48s (x16 over 2m6s)  kubelet            spec.containers{time
outs}: Readiness probe failed: HTTP probe failed with statuscode: 404
  Normal   Killing    33s (x3 over 2m3s)   kubelet            spec.containers{time
outs}: Container timeouts failed liveness probe, will be restarted
  Normal   Pulled     3s (x4 over 2m17s)   kubelet            spec.containers{time
outs}: Container image "nginx:1.25" already present on machine and can be accessed
 by the pod
  Normal   Created    2s (x4 over 2m17s)   kubelet            spec.containers{time
outs}: Container created
  Normal   Started    2s (x4 over 2m17s)   kubelet            spec.containers{time
outs}: Container started
```

describe

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: bad-ports
  labels:
    app: bad-ports
spec:
  selector:
    matchLabels:
      app: bad-ports
  template:
    metadata:
      labels:
        app: bad-ports
    spec:
      containers:
      - name: bad-ports
        image: redis:7-alpine
        ports:
        - containerPort: 6379

        livenessProbe:
          tcpSocket:
            port: 6380
          initialDelaySeconds: 10
          periodSeconds: 5

```

ports.yaml

```
lly@lly ~/m/s/t/4 (master)> kubectl get pods
NAME                          READY   STATUS    RESTARTS      AGE
bad-ports-6b4649bcfc-ltcdq    1/1     Running   3 (8s ago)    69s
```

kubectl get pods

```
Events:
  Type     Reason     Age                From               Message
  ----     ------     ----               ----               -------
  Normal   Scheduled  86s                default-scheduler  Successfully assigned
default/bad-ports-6b4649bcfc-ltcdq to minikube
  Normal   Pulled     5s (x5 over 85s)   kubelet            spec.containers{bad-po
rts}: Container image "redis:7-alpine" already present on machine and can be acces
sed by the pod
  Normal   Created    5s (x5 over 85s)   kubelet            spec.containers{bad-po
rts}: Container created
  Normal   Started    5s (x5 over 85s)   kubelet            spec.containers{bad-po
rts}: Container started
  Warning  Unhealthy  5s (x12 over 75s)  kubelet            spec.containers{bad-po
rts}: Liveness probe failed: dial tcp 10.244.0.55:6380: connect: connection refuse
d
  Normal   Killing    5s (x4 over 65s)   kubelet            spec.containers{bad-po
rts}: Container bad-ports failed liveness probe, will be restarted
```

kubectl describe <pod_name>

```
lly@lly ~/m/s/t/4 (master)> kubectl logs bad-ports-6b4649bcfc-ltcdq
1:C 31 Mar 2026 14:41:36.720 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 31 Mar 2026 14:41:36.720 * Redis version=7.4.8, bits=64, commit=00000000, modi
fied=0, pid=1, just started
1:C 31 Mar 2026 14:41:36.720 # Warning: no config file specified, using the defaul
t config. In order to specify a config file use redis-server /path/to/redis.conf
1:M 31 Mar 2026 14:41:36.720 * monotonic clock: POSIX clock_gettime
1:M 31 Mar 2026 14:41:36.721 * Running mode=standalone, port=6379.
1:M 31 Mar 2026 14:41:36.721 * Server initialized
1:M 31 Mar 2026 14:41:36.721 * Ready to accept connections tcp
1:signal-handler (1774968116) Received SIGTERM scheduling shutdown...
1:M 31 Mar 2026 14:41:56.399 * User requested shutdown...
1:M 31 Mar 2026 14:41:56.399 * Saving the final RDB snapshot before exiting.
1:M 31 Mar 2026 14:41:56.413 * DB saved on disk
1:M 31 Mar 2026 14:41:56.413 # Redis is now ready to exit, bye bye...
```

kubectl logs <pod_name>

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: long-launch
  labels:
    app: long-launch
spec:
  selector:
    matchLabels:
      app: long-launch
  template:
    metadata:
      labels:
        app: long-launch
    spec:
      containers:
      - name: long-launch
        image: alpine
        command: ["/bin/sh"]
        args: ["-c", "sleep 10;"]
        ports:
        - containerPort: 80

        livenessProbe:
          httpGet:
            path: /
            port: 80
          initialDelaySeconds: 2
          periodSeconds: 5

```

long.yaml

```
lly@lly ~/m/s/t/4 (master)> kubectl apply -f long.yaml
deployment.apps/long-launch created
lly@lly ~/m/s/t/4 (master)> kubectl get pods -w
NAME                              READY   STATUS              RESTARTS   AGE
long-launch-5b94855f5b-5ctnz      0/1     ContainerCreating   0          2s
long-launch-5b94855f5b-5ctnz      1/1     Running             0          4s
long-launch-5b94855f5b-5ctnz      0/1     Completed           0          14s
long-launch-5b94855f5b-5ctnz      1/1     Running             1 (4s ago)   16s
long-launch-5b94855f5b-5ctnz      1/1     Running             2 (2s ago)   27s
long-launch-5b94855f5b-5ctnz      0/1     Completed           2 (12s ago)   37s
long-launch-5b94855f5b-5ctnz      0/1     CrashLoopBackOff    2 (4s ago)    40s
```

kubectl get pods -w (приложение переходит в состояние running за две секунды, через 10 секунд, после выполнения команды sleep 10, оно перезапустится)

Контрольные вопросы для всез заданий

В чем основное отличие LivenessProbe от ReadinessProbe? – livenessProbe проверяет “жив” ли контейнер и если нет, то перезапустить его. ReadinessProbe определяет готовность принимать трафик.

Какие типы проверок поддерживают probes и когда каждый из них следует использовать? – exec, httpGet, tcpSocket. Exec – для проверки через выполнение команды, httpGet – через http запрос, tcpSocket – проверка через подключение

Что произойдет, если не настроить readinessProbe для приложения с долгим стартом? – приложение будет считаться готовым и могут появиться ошибки

Какой смысл параметров initialDelaySeconds, periodSeconds, timeoutSeconds? Что означает failureThreshhold и successThreshold? – iniditalDelaySeconds – время от старта контейнера до запуска livenessProbe, periodSeconds – кол-во секунд между пробами, timeoutSeconds – кол-во секунд ожидания пробы. FailureThreshold - сколько раз произвести команду, прежде чем считать контейнер мертвым, SuccessThreshold – кол-во проверок, чтобы проба считалась неудачной.

Какие три команды вы бы использовали для диагностики проблем с livenessProbe – describe, logs, get pods

Что делать если контейнер постоянно перезапускается (CrashLoopBackOff)? Посмотреть логи, увеличить initialDelaySeconds

Как узнать, сколько раз контейнер пекрезапускался из за провала livenessProbe? – get pods

Какие события в kubectl describe pod указывают на проблемы с probes? – warning unhealthy

Как проверить, что ReadinessProbe корректно исключает под из балансировки? – отсутсвие endpoint’а или под не будет в состоянии ready.

Как доказать, что два контейнера используют одинаковый физический файл? – у них будет одинаковый inode

Какой тип probe (Liveness/Readiness) должне сработать, если ваше приложение полностью функционирует, но временно может не обрабатывать новые запросы из за перезагрузки базы данных? – readiness

Что произойдет с данными в emptyDir, если узел (node), на котором работает pod, перезагрузится? – они потеряются